

AI-Based Forecasting to Shift Epidemiological Surveillance Toward Predictive Approaches and Inform Health System Planning (EpiForecast-)

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Background and aims:

- Epidemiological surveillance systems are still widely used for retrospective and descriptive purposes. There is a growing need to shift surveillance toward predictive approaches
- More than 19 million Mexicans live with some form of mental illness I, dementia or Parkinson's disease
- The Mexican National Systems of Surveillance collects epidemiological data weekly. Epidemiological surveillance systems are still widely used for retrospective and descriptive purposes
- But there is no system to turn them into forecasts for planning ahead
- Public health is shifting from retrospective surveys toward predictions based on longitudinal real-time records.
- Epidemiological intelligence now emerges from real operational data, not only statistical inference

Objective:

- In this study, neuropsychiatric disorders—such as depression, Alzheimer's disease, and Parkinson's disease—are used as practical examples to develop and evaluate an artificial intelligence (AI)-based framework aimed at forecasting weekly incidence trends

- While the framework is applied using national surveillance data from Mexico, its design is intended to be adaptable and scalable to other health systems with similar data structures

Methods:

- A retrospective observational study using national surveillance data from 2014–2026, including 333 time series at state, regional, and national levels
- First stage: Four predictive models (Prophet, DeepAR, XGBoost ensemble, and Ridge stacking) were developed to capture temporal dynamics and variability across conditions
- Second stage: Models were evaluated using expanding-window cross-validation to ensure temporal generalizability. Performance was assessed using mean absolute scaled error (MASE) and symmetric mean absolute percentage error (SMAPE), selecting the best-performing model for each series

Results:

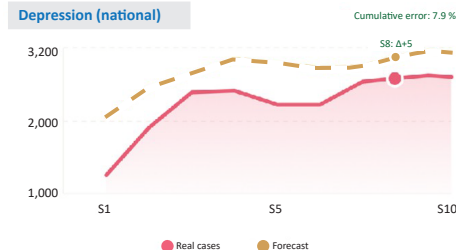
Depression showed the highest predictive accuracy (median MASE=0.17; SMAPE=5.99%), outperforming baseline models in 97.3% of series and capturing a ~12% annual increase after COVID-19. Parkinson's disease showed moderate performance (MASE=0.26; SMAPE=35.24%), while Alzheimer's disease showed higher relative error due to low incidence but low absolute deviation (median MAE=0.6 cases/week). DeepAR was selected in 73.3% of series. Forecasts revealed spatial and temporal variations, enabling early identification of potential service demand pressures.

Conclusions:

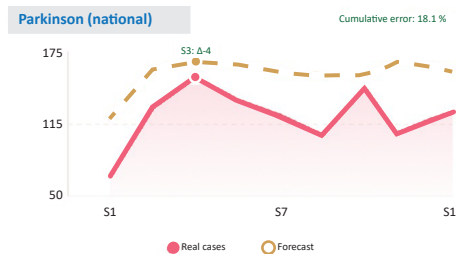
- These findings highlight how integrating AI-based forecasting into epidemiological surveillance can support a shift from descriptive monitoring toward predictive decision-making
- By identifying future demand patterns and geographic hotspots, this approach can improve resource allocation and reduce uncertainty in planning
- This framework represents a feasible and transferable step toward more responsive, data-driven, and forward-looking health systems

EpiForecast-

Depression (national)



Parkinson (national)



Alzheimer (national)

